Rational and irrational numbers

1 Classify the following numbers as rational or irrational:

a 4 b -4 c 4.435 d $\frac{3}{13}$

e $\frac{7}{6}$ f -0.7 g π h 0

2 Is $\sqrt{15}$ an integer?

3 Classify the following numbers as rational or irrational:

a $\sqrt{7}$ b $\left(\sqrt{7}\right)^2$ c $\sqrt{35}$ d $\sqrt{81}$

e $\sqrt{225}$ f $-\sqrt{13}$ g $-\sqrt{64}$ h $\sqrt[3]{23}$

i $\sqrt[3]{64}$ j $\sqrt[3]{512}$

4 State whether the following expressions simplify into a rational or an irrational number:

a π^2 b $\sqrt{\pi^2}$ c $\sqrt{9}$ d $\sqrt{e} \times \sqrt{e}$

e $\sqrt{\frac{\pi}{\pi}}$ f $\frac{\pi\sqrt{\pi}}{\sqrt{\pi}}$ g $\frac{5\sqrt{5}}{\sqrt{5}}$ h $\frac{\sqrt{e}}{\sqrt{e}}$

5 Evaluate the following, rounding your answers to two decimal places where necessary:

a $\sqrt{25}$ b $\sqrt[3]{64}$ c $\sqrt{33}$ d $\sqrt[3]{26}$

e $\sqrt{121}$ f $\sqrt[3]{216}$ g $\sqrt{83}$ h $\sqrt[3]{52}$

Operations with rational and irrational numbers

6 State whether the sum of the following numbers is a rational or an irrational number:

a $\frac{3}{10} + \frac{7}{8}$ b $-\frac{2}{7} + \frac{1}{11}$ c $\frac{4}{7} + \left(-\frac{9}{11}\right)$ d $\frac{3}{11} + \sqrt{7}$

e $\sqrt{11} + \frac{4}{7}$ f $\frac{5}{11} + \left(-\sqrt{5}\right)$ g $\sqrt{11} + \left(-\frac{3}{5}\right)$ h $-\frac{3}{5} + \sqrt{4}$

7 State whether the product of the following [redacted]ers is a rational or an irrational number:

a $\frac{1}{5} \times \frac{1}{9}$ **b** $-\frac{3}{7} \times \frac{9}{10}$ **c** $0 \times (-\sqrt{2})$ **d** $\sqrt{6} \times (-\sqrt{6})$

e $\frac{3}{11} \times \sqrt{7}$ **f** $\sqrt{5} \times \frac{9}{11}$ **g** $-\frac{4}{11} \times \sqrt{7}$ **h** $\sqrt{3} \times \sqrt{6}$

8 State whether the following will result to a rational number, irrational number or either:

a The sum of two rational numbers.

b The product of two rational numbers.

c The sum of two irrational numbers.

d The product of two irrational numbers.

e The sum of a rational and an irrational number.

f The product of two non-zero rational numbers.

g The product of a non-zero rational number and an irrational number.

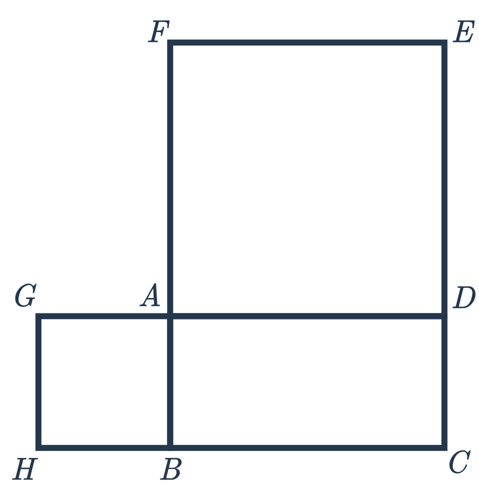
9 Let f and g be integers that satisfy the following conditions:

- $1 < f < g < 10$
- $\sqrt{f} + \sqrt{g}$ is irrational.
- $\sqrt{f} \times \sqrt{g}$ is rational.

State whether the following values of f and g satisfies the given conditions:

a $f = 2,\ g = 8$ **b** $f = 3,\ g = 12$ **c** $f = 2,\ g = 4$ **d** $f = 8,\ g = 2$

10 In the given diagram, a rectangle is formed using the side lengths of two different squares. $ADEF$ is a square with area measuring 63 square units, and $ABHG$ is a square with area measuring 49 square units:



a Write an expression that represents the area of rectangle $ABCD$.

b Is the area of rectangle $ABCD$ a rational or irrational value?